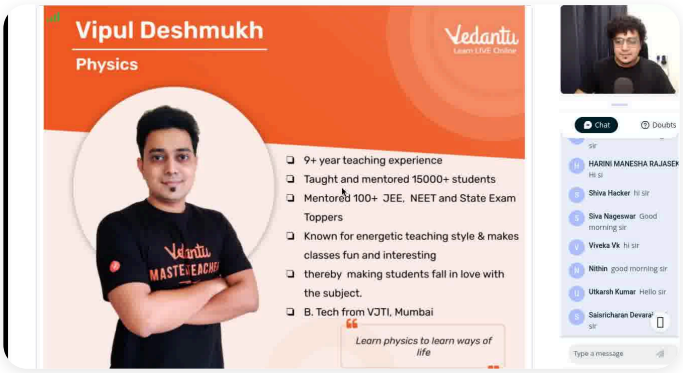
Notes

Introduction to Electrostatics



Welcome and Instructor Introduction

Hello everyone! My name is Vipul Deshmukh, and I will be your master teacher for the Class 12th chapters in this course. I have 9+ years of teaching experience, have mentored over 15,000 students, and am known for an energetic teaching style that makes physics fun and interesting. I hold a B.Tech from VJTI, Mumbai.

Chapter Overview: Electrostatics

We are starting with the chapter **Electrostatics**. This is a very important chapter, and it is closely related to gravitation. Understanding electrostatics thoroughly will make gravitation easier to grasp, and vice-versa.

Course Flow

Our course will follow a modified flow, different from the NCERT sequence, as we are preparing purely for entrance exams and not subjective board exams. The sequence will be:

1. **Electrostatics**
2. **Current Electricity**
3. **Capacitors**

This flow is chosen because questions involving networks of resistors in current electricity and capacitors are often identical in their problem-solving approach. We will begin by discussing charges.

What is Electrostatics?

Electrostatics is a branch of physics that deals with the study of electric charges at rest. The term "electro" refers to electric charge (derived from the word "electron"), and "statics" means at rest. In this chapter, we will focus exclusively on the behavior and effects of charges that are stationary.

- **Electrostatics:** Deals with charges that are *at rest*. The effects observed are due to stationary charges.
- **Electrodynamics:** Deals with charges that are *in motion*. This field is commonly known as current electricity, where charges flow through conductors, creating electric currents. We will study electrodynamics in later chapters.

Scope of Electrostatics

In electrostatics, we will explore all phenomena and effects associated with a charge that is not moving or flowing. This includes understanding concepts such as electric fields, electric potential, and forces between stationary charges.

Origin of Charge and Atomic Structure

True Origin of Charge

The true origin of charge is rooted in the structure of the atom.

Atomic Structure

An atom consists of:

- **Nucleus:** Located at the center, containing:
  - **Neutrons:** Electrically neutral particles, possessing no charge.
  - **Protons:** Positively charged particles.
- **Electrons:** Negatively charged particles that revolve around the nucleus.

Types of Charge and Convention

- There are two different kinds of charges, distinguished by convention:
  - **Proton Charge:** Designated as positive ($+e$).
  - **Electron Charge:** Designated as negative ($-e$).

Value and Unit of Elementary Charge

- The magnitude of charge on both a proton and an electron is exactly the same.
- **Value:** 1.6×10^{-19} Coulomb.
- **Unit:** Coulomb (C), named after Coulomb, who formulated Coulomb's Law.

Charge of a Neutral Atom

- A neutral atom (not an ion) always contains an equal number of electrons and protons.

Origin of Charge

Understanding Charge

When a body is said to have a charge, it implies an imbalance between the number of protons and electrons within that body. Protons carry a positive charge, and electrons carry an equal but opposite negative charge.

Neutral Objects

An object is considered neutral when the number of protons is equal to the number of electrons. In this state, the positive charges from protons cancel out the negative charges from electrons, resulting in a net charge of zero.

Charging by Adding Electrons

If extra electrons are added to a neutral body, an imbalance is created. For example:

- **Adding one electron:** If a body initially has 10 protons and 10 electrons (net charge = 0), adding one electron results in 10 protons and 11 electrons. The 10 protons are canceled by 10 electrons, leaving one extra electron. The body's charge becomes negative, equal to the charge of one electron, which is $-e$ or -1.6×10^{-19} C.
- **Adding two electrons:** If two extra electrons are added, the body will have 10 protons and 12 electrons. After cancellation, two extra electrons remain. The body's charge becomes $-2e$, or -3.2×10^{-19} C.
- **Generalizing:** If 'n' extra electrons are added, the charge on the body will be $-ne$.

Charging by Removing Electrons

Similarly, if electrons are removed from a neutral body, a positive charge is created:

- **Removing one electron:** If a body initially has 10 protons and 10 electrons, removing one electron leaves 10 protons and 9 electrons. The 9 electrons cancel 9 protons, leaving one extra proton. The body's charge becomes positive, equal to the charge of one proton, which is $+e$ or $+1.6 \times 10^{-19}$ C.
- **Removing two electrons:** If two electrons are removed, the body will have 10 protons and 8 electrons. After cancellation, two extra protons remain. The body's charge becomes $+2e$, or $+3.2 \times 10^{-19}$ C.
- **Generalizing:** If 'n' electrons are removed, the charge on the body will be $+ne$.

Key Principle: Electron Movement

deficiency (less number) or an abundance (more number) of electrons.

Quantization of Charge

Definition

Quantization of charge states that charge cannot have any random value. Instead, it exists in discrete packets or multiples of a fundamental unit of charge.

Reason for Quantization

The charge on a body is due to the excess or deficiency of electrons. Since electrons are fundamental particles and cannot be divided, charge can only be transferred in integer multiples of the charge of a single electron.

Mathematical Representation

The magnitude of charge (Q) on any object is given by: $Q = \pm ne$ Where:

- n is an integer (1, 2, 3, ...), representing the number of electrons transferred.
- e is the elementary charge, which is the magnitude of the charge of a single electron or proton. Its value is approximately 1.6×10^{-19} Coulombs.

Implications

- You can have a charge of $1e$, $2e$, $3e$, etc., but not $1.5e$ or $2.7e$.
- For example, possible charge values could be 1.6×10^{-19} C (for $n = 1$) or 3.2×10^{-19} C (for $n = 2$). Values between these are not possible.

Sign Convention

- If the charge is positive, it means the body has a deficiency of n electrons.
- If the charge is negative, it means the body has an excess of n electrons (i.e., more electrons than protons).

Real-Life Analogy

Consider collecting a donation of 16 rupees from each student in a class:

- If there are 10 students, the total collected is 10×16 rupees.
- If there are 9 students, the total collected is 9×16 rupees.
- The total amount collected will always be a multiple of 16 rupees (e.g., 1×16, 2×16, 3×16, etc.). It's not possible to collect an amount like 1.5×16 rupees, because you cannot have half

Quantization of Charge: Calculating Number of Electrons

Quantization Formula

The magnitude of charge (Q) is given by the formula: $Q = nE$ Where:

- Q is the total charge.
- n is the number of electrons (an integer).
- E is the elementary charge, which is approximately 1.6×10^{-19} Coulombs.

Example 1: Positive Charge

Question: Suppose we are given a charge of $+8$ Coulombs. Find the number of electrons associated with this charge.

Solution:

1. **Given:** $Q = +8$ C
2. **Formula:** $Q = nE$
3. **Rearrange for n:** $n = \frac{Q}{E}$
4. **Substitute values:** $n = \frac{8}{1.6 \times 10^{-19}}$
5. **Calculation:** $n = \frac{80}{16} \times 10^{19}$ $n = 5 \times 10^{19}$

Result: The number of electrons is 5×10^{19}.

Interpretation: Since the charge is positive, it indicates a deficiency of electrons. Therefore, 5×10^{19} electrons are less than the neutral state.

- For a positive charge, electrons are less in number (deficiency).
- For a negative charge, electrons are more in number (abundance).

Example 2: Impossible Charge Value

Question: Consider a charge of -8×10^{-20} Coulombs. Is this charge possible? If so, what is the number of electrons associated?

Solution:

1. **Given:** $Q = -8 \times 10^{-20}$ C
2. **Formula:** $Q = nE$
3. **Rearrange for n:** $n = \frac{|Q|}{E}$ (We use the magnitude of charge for calculation of 'n')
4. **Substitute values:** $n = \frac{8 \times 10^{-20}}{1.6 \times 10^{-19}}$

Result: The calculated value for n is 0.5.

Interpretation: According to the principle of quantization of charge, the number of electrons (n) must be an integer. Since 0.5 is not an integer, this type of charge (-8×10^{-20} C) is not possible. It is impossible to have half an electron.

Convention of Charge Signs

It's important to remember that positive and negative signs for charge are merely conventions to distinguish between different types of charge (e.g., proton vs. electron). It does not imply that one charge is numerically 'less' or 'more' than the other in terms of magnitude, only in terms of its nature.

Quiz: Electrons in One Coulomb Charge

Question: What will be the number of electrons in one Coulomb charge?

Solution:

1. **Given:** $Q = 1$ C
2. **Formula:** $Q = nE$
3. **Rearrange for n:** $n = \frac{Q}{E}$
4. **Substitute values:** $n = \frac{1}{1.6 \times 10^{-19}}$
5. **Calculation:** $n = \frac{1 \times 10^{19}}{1.6}$ $n = \frac{10 \times 10^{18}}{1.6}$ $n = 6.25 \times 10^{18}$

Result: The number of electrons is 6.25×10^{18}.

Interpretation: This is a very large number, indicating that one Coulomb is a significant amount of charge. If this is a positive charge, then 6.25×10^{18} electrons are less. If it is a negative charge, then 6.25×10^{18} electrons are more.

Effect of Charging on Mass

Understanding Charge and Mass

When a body is charged, it means there is an imbalance between the number of protons and electrons. This imbalance is achieved by either adding or removing electrons.

Mass of an Electron

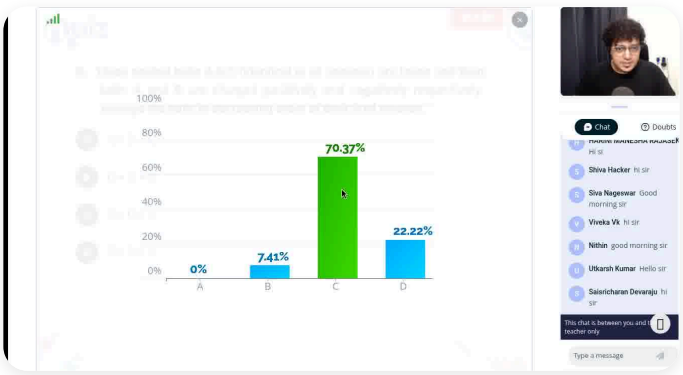
An electron possesses a very small, but non-zero, mass. The mass of a single electron is approximately 9.1×10^{-31} kg.

mass of the body will slightly **decrease**. This is because each electron removed carries away its tiny mass.

- **Adding Electrons:** If electrons are added to a body to give it a negative charge, the total mass of the body will slightly **increase**. This is due to the added mass of the electrons.

Although the mass of a single electron is extremely small, when a significant quantity of electrons is transferred, the change in mass can become measurable and significant in certain contexts.

Quiz: Mass Change Due to Charging



Problem Statement

Three neutral balls, A, B, and C, are identical in all aspects. Ball A is given a positive charge, ball B is given a negative charge, and ball C remains neutral. Arrange the balls in decreasing order of their final mass.

Solution Explanation

To understand the change in mass, we need to recall how objects acquire positive and negative charges:

- **Negative Charge:** An object acquires a negative charge by gaining electrons. Since electrons have mass, adding electrons to an object increases its total mass.
- **Positive Charge:** An object acquires a positive charge by losing electrons. Removing electrons from an object decreases its total mass.
- **Neutral Charge:** A neutral object neither gains nor loses electrons, so its mass remains unchanged.

Let's apply this to the given balls:

1. **Ball B (Negative Charge):** Ball B is given a negative charge, which means electrons are added to it. Therefore, the mass of ball B will **increase**.
2. **Ball C (Neutral Charge):** Ball C remains neutral, meaning no electrons are added or removed. Therefore, its mass will **remain the same** as its initial mass.

Comparing the final masses:

- Mass of B > Initial Mass
- Mass of C = Initial Mass
- Mass of A < Initial Mass

Arranging them in decreasing order of their final mass:

$$\text{Mass of B} > \text{Mass of C} > \text{Mass of A}$$

Thus, the correct order is B, C, A.

Final Answer

The correct option is C, which corresponds to the order B > C > A.

Properties of Charge: Additivity

Additivity of Charges

Charges can be added or subtracted algebraically, considering their signs. This means that the basic rules of algebra apply to charges.

For example, if a body has multiple charges at different points, the total charge is the algebraic sum of all individual charges.

Consider a plastic piece with the following charges:

- +5 C
- −7 C
- −8 C
- +2 C

To find the total charge, we perform an algebraic sum: $\text{Total Charge} = (+5\text{ C}) + (-7\text{ C}) + (-8\text{ C}) + (+2\text{ C})$ $\text{Total Charge} = (5 + 2)\text{ C} + (-7 - 8)\text{ C}$ $\text{Total Charge} = 7\text{ C} - 15\text{ C}$ $\text{Total Charge} = -8\text{ C}$

Thus, the total charge on the system is -8 C.

Properties of Charge: Conservation of Charge

Conservation of Charge

In an isolated system, the **total charge remains constant** (conserved). While charges within the system may redistribute or exchange amongst themselves, their algebraic sum will always stay the same.

Example: Consider an isolated system with a sphere and a rod.

- **Initial State:**
  - Sphere: +12 C
  - Rod: 0 C (neutral)
  - Total initial charge: 12 C + 0 C = 12 C
- **Process:** The sphere and the rod are brought into contact with each other, allowing charge transfer.
- **Final State:**
  - Rod: +7 C
  - Sphere: Q_{sphere} (unknown)

According to the principle of conservation of charge, the total charge in the isolated system must remain constant.

Total Initial Charge = Total Final Charge 12 C = Q_{sphere} + 7 C Q_{sphere} = 12 C − 7 C Q_{sphere} = 5 C

Therefore, the final charge on the sphere is +5 C. The total final charge is 5 C + 7 C = 12 C, which is equal to the total initial charge.

Auxiliary Properties of Charge

Charge is Independent of State of Matter

Charge does not depend on the state of matter (rest or motion). If a body has a charge of +5 Coulomb, it will remain +5 Coulomb whether the body is at rest or in motion. The charge value remains constant regardless of its state of motion.

Charge Cannot Exist Without Matter

Charge is always associated with matter. The origin of charge is due to an abundance or deficiency of electrons within matter. If there is no matter, there can be no excess or deficiency of electrons, and thus no charge. For example, in a vacuum, we cannot have charge because there is no matter

Notes

Interaction Between Charges

Like charges repel each other, and unlike charges attract each other. This is a fundamental property of charges:

- **Like Charges:**
  - Positive (+) and Positive (+) charges repel.
  - Negative (-) and Negative (-) charges repel.
- **Unlike Charges:**
  - Positive (+) and Negative (-) charges attract.

If one object is charged (e.g., positive) and another is neutral, there will generally be no significant interaction if the charges are small. These are some of the key properties of electric charges.

Problem: Attraction and Repulsion of Charged Bodies

Question

Two charged bodies P and Q attract each other. Two bodies Q and R repel each other. If P and R are held together, what will happen?

Solution

This problem relies on the fundamental principle that like charges repel and unlike charges attract. We can assume the nature of the charges for P, Q, and R to determine their interaction.

Case 1: Assuming P is positive

1. **P and Q attract each other:** If P is positive ($P = +$), then for attraction to occur, Q must be negative ($Q = -$).
2. **Q and R repel each other:** Since Q is negative ($Q = -$), for repulsion to occur, R must also be negative ($R = -$).
3. **P and R held together:** In this scenario, P is positive ($P = +$) and R is negative ($R = -$). Since they have opposite charges, P and R will attract each other.

Case 2: Assuming P is negative

1. **P and Q attract each other:** If P is negative ($P = -$), then for attraction to occur, Q must be positive ($Q = +$).
2. **Q and R repel each other:** Since Q is positive ($Q = +$), for repulsion to occur, R must also be positive ($R = +$).
3. **P and R held together:** In this scenario, P is negative ($P = -$) and R is positive ($R = +$). Since they have opposite charges, P and R will attract each other.

Conclusion

Regardless of the initial charge assumption, P and R will attract each other.

Charging of a Conductor

Introduction to Charging Methods

So far, we have discussed the fundamental properties of charges, including how to impart positive or negative charges to an object by removing or adding electrons. Now, we will delve into the practical procedures and different methods used to charge a conductor. This section focuses on the various techniques for charging, building upon our understanding of charge properties and related questions.

Conductors and Insulators

Introduction to Conductors and Insulators

To understand charging methods, it's crucial to differentiate between various types of materials based on their ability to conduct electricity.

Conductors

- **Definition:** Materials that allow electric charge (flow of electrons) to pass through them easily.
- **Microscopic View:** In conductors, especially metals, there are numerous "free electrons" available. These electrons are not tightly bound to individual atoms and can move freely within the material under suitable conditions.
- **Nuclear Hold:** The nucleus in a conductor has a very weak hold or no control over the electrons in the outermost orbit. These electrons can easily jump out of their orbits and become mobile.
- **Examples:** Metals (e.g., copper wire), Earth.
- **Current Flow:** Since free electrons are available and can move, current can flow through conductors.

Insulators (Non-Conductors)

- **Definition:** Materials that do not allow electric charge to pass through them easily. Electrons are not allowed to move freely within these materials.
- **Microscopic View:** In insulators, electrons are tightly bound to the nucleus of their atoms. There are no free electrons available for conduction.
- **Nuclear Hold:** The nucleus in an insulator has a strong hold on its electrons, especially those in the outermost orbit. They are very tightly packed and cannot easily jump out or move away

- **Current Flow:** Due to the absence of free electrons, current cannot travel through insulators.

Key Difference: Nuclear Hold on Electrons

- **Conductors:** Nucleus has a weak or no hold on outermost electrons, allowing them to become free and mobile.
- **Insulators:** Nucleus has a strong hold on all electrons, preventing them from becoming free and mobile.

Practical Applications and Safety

- **Wood:** A good example of a non-conductor. In case of electric shock, a wooden stick is used to separate the person from the live wire because it won't conduct electricity.
- **Rubber/Plastic:** Electricians use rubber gloves and wear rubber slippers (like Hawaiian slippers) because these materials are insulators, providing protection against electric shock.

Temperature Dependency

The dependency of temperature on conductance is a more advanced topic that will be covered in the chapter "Current Electricity." For now, the focus is on the basic distinction between conductors and insulators.

Methods of Charging

Introduction to Charging Methods

There are various methods for charging objects, which can be broadly categorized based on the type of material being charged:

- **Charging by Friction:** This method is exclusively applicable to **non-conductors** (insulators).
- **Charging by Conduction:** This method is applicable only for **conductors**.
- **Charging by Induction:** This method is also applicable only for **conductors**.

Effect of Temperature on Conductors

The effect of temperature on conductors will be discussed in detail in the subsequent chapter on "Current Electricity."

Charging by Friction



Principle of Charging by Friction

Charging by friction is a simple method primarily used for insulators. When two non-conducting objects are rubbed against each other, friction generates heat. This heat provides the energy for electrons to transfer from one object to the other, resulting in both objects becoming charged.

Electron Transfer in Friction

The direction of electron transfer (which object loses electrons and which gains them) depends on the electron affinity of the materials involved. The object with a weaker attraction for its electrons will tend to lose them, becoming positively charged. Conversely, the object with a stronger electron affinity will gain electrons, becoming negatively charged.

Example: Glass Rod and Silk Cloth

When a glass rod is rubbed against a silk cloth:

- The glass rod has less attraction for its electrons, so it **loses electrons** and becomes **positively charged**.
- The silk cloth has a higher affinity for electrons, so it **gains electrons** and becomes **negatively charged**.

Other Examples of Charging by Friction

- **Plastic Comb and Dry Hair:** Rubbing a plastic comb through dry hair causes the comb to become charged. This charged comb can then attract small pieces of paper or even lift strands of hair due to static electricity. It's crucial for the hair to be dry (not oily or wet) because oil or water would act as a lubricant, reducing friction and hindering charge transfer.
- **Plastic Ruler and Water Stream:** A plastic ruler rubbed on hair can be used to deflect a tiny stream of water. The charged ruler attracts the polar water molecules, causing the stream to bend.

Charging by Conduction

Definition

Charging by conduction is a method of charging an object that is applicable only to **conductors**. It involves making direct physical contact between a charged object and an uncharged object.

uncharged object (e.g., a neutral sphere).

2. **Contact:** The charged object is brought into physical contact with the uncharged object.
3. **Charge Transfer:** Upon contact, some of the charges from the heavily charged object transfer to the uncharged object.
4. **Separation:** The charged object is then removed.
5. **Final State:** Both objects now possess a charge. The original charged object will have some charge remaining, and the previously uncharged object will acquire a charge of the same sign.

Key Principles

- **Physical Contact is Essential:** Unlike induction, conduction requires direct touch.
- **Charge Distribution:** The charge will not always be equally distributed between the two objects. The distribution depends on the geometry (shape) of the objects.
- **Equipotential:** The core logic behind charge transfer in conduction is that charge will continue to transfer until both objects reach the **same electric potential**.

Microscopic Explanation

Consider a rod with a large number of negative charges. These negative charges repel each other and try to move as far away from each other as possible. They are initially confined to the surface of the rod.

When this charged rod touches a neutral conductor (like a sphere), it provides an additional, empty space (surface area) for these repelling charges. The electrons, driven by their mutual repulsion, seize this opportunity to spread out over the combined surface area of both objects. This movement of electrons from the rod to the sphere continues until the repulsive forces are balanced, and both objects achieve the same potential.

Analogy: The Bus Stop

Imagine a crowded bus (Bus 1) full of people (representing negative charges) who are uncomfortable due to being packed closely together (mutual repulsion). This bus is waiting at a stop.

Suddenly, an empty bus (Bus 2, representing the neutral sphere) arrives at the same stop, going to the same destination. What happens?

- Some people from the crowded Bus 1, especially those standing, will move to the empty Bus 2 to find a seat and more space.
- Not everyone will move; only enough people will transfer so that the discomfort (repulsion) is reduced, and people can spread out more comfortably across both buses.

Similarly, in charging by conduction, the electrons from the highly charged object (Bus 1) move to the neutral conductor (Bus 2) to maximize their separation due to repulsive forces, until both objects reach an equilibrium state (same potential).

Charging by Induction

Introduction to Charging by Induction

Charging by induction is a method of charging an object without making physical contact with the charging source. This process relies on the principle of electrostatic attraction or repulsion.

Process of Charging by Induction

1. **Bringing a Charged Object Near:** A charged object (e.g., a rod) is brought near a neutral conductor (e.g., two spheres in contact).
2. **Charge Separation:** The electric field of the charged object causes the free charges within the neutral conductor to redistribute. For instance, if a positively charged rod is brought near, it attracts negative charges (electrons) towards it and repels positive charges to the opposite side of the conductor.
3. **Separation of Conductors:** While the charged object is still held near, the conductors are separated. This traps the separated charges on each conductor.
4. **Removal of Charged Object:** Once the conductors are separated, the charged object can be removed. The separated charges remain on the conductors, resulting in two oppositely charged objects.

Key Characteristics of Induction:

- **No Physical Contact:** The charging object does not touch the object being charged.
- **No Charge Transfer from Source:** There is no actual transfer of charge from the charging object to the object being charged. The charging object merely influences the redistribution of charges within the neutral object.
- **Nature of Objects:** The object being charged *must* be a conductor. The charging object (e.g., the rod) may or may not be a conductor, as its role is to create an electric field, not to transfer charge.

Example Scenario:

Consider two neutral conducting spheres in contact. If a positively charged rod is brought near one sphere:

- Electrons from both spheres will be attracted towards the rod, accumulating on the sphere closer to the rod.
- The sphere further away will be left with an excess of positive charges.

Permanence of Charging by Induction

The permanence of the charge depends on the completion of the process:

- If the conductors are separated while the inducing charged object is still nearby, the charging is sustained and permanent.
- If the inducing charged object is removed *before* the conductors are separated, the charges will redistribute back to their original neutral state, and the objects will regain their original structure.

Comparison with Other Charging Methods

1. **Charging by Friction:**
   - Applicable only for non-conductors (insulators).
   - Involves rubbing two objects together.
   - Electrons are transferred from the object with less electron affinity to the one with higher electron affinity, resulting in one positively charged and one negatively charged object.
2. **Charging by Conduction:**
   - Requires two objects, both of which must be conductors.
   - One object is heavily charged, and the other is neutral.
   - Physical contact is made between the charged and neutral objects.
   - Charge transfers from the heavily charged object to the neutral object until charge distribution is achieved.
   - The total charge of the system remains constant; the charge lost by one object is gained by the other.
3. **Charging by Induction:**
   - Influences charge distribution without physical contact.
   - Relies on attraction or repulsion to achieve charge separation.
   - The object being charged must be a conductor.

Coulomb's Law

Introduction to Coulomb's Law

Coulomb's Law describes the force between two point charges. Although the law is often illustrated with larger objects like spheres, its fundamental application is for point charges.

Analogy to Newton's Law of Gravitation

Coulomb's Law is very similar in its mathematical form to Newton's Law of Gravitation.

- **Like charges repel:** Charges with the same polarity (both positive or both negative) push each other away.
- **Unlike charges attract:** Charges with opposite polarities (one positive and one negative) pull each other closer.

Factors Affecting the Magnitude of Force

The magnitude of the electrostatic force is determined by two main factors:

1. **Direct Proportionality to the Product of Charges:**
    - The force is directly proportional to the product of the magnitudes of the two charges (Q_1 and Q_2).
    - This means that if the charges are larger in magnitude, the attractive or repulsive force between them will be stronger. Conversely, smaller charges result in weaker forces.
    - Mathematically, this can be expressed as: $F \propto Q_1 Q_2$
2. **Inverse Proportionality to the Square of the Distance:**
    - The force is inversely proportional to the square of the distance (r) between the two charges.
    - This is known as the **inverse square law**, similar to gravitation.
    - If the charges are closer together, the force between them will be significantly stronger.
    - If the charges are further apart, the force will decrease rapidly.
    - Mathematically, this can be expressed as: $F \propto \frac{1}{r^2}$

Coulomb's Constant and Permittivity

Coulomb's Constant (k)

When converting the proportionality in Coulomb's Law to an equality, a constant, denoted as small 'k', is introduced. This 'k' is known as Coulomb's constant.

Mathematically, Coulomb's constant is defined as: $k = \frac{1}{4\pi\epsilon_0}$

Permittivity (ϵ)

- **Symbol:** The Greek letter epsilon (ϵ) is used to represent permittivity.
- **Definition:** Permittivity is a property of the medium in which the charges are placed.
- **Medium Dependence:** Permittivity varies for different mediums (e.g., air, water, kerosene). Each material has its own unique permittivity.

Permittivity of Free Space (ϵ_0)

- **Symbol:** The permittivity of air, vacuum, or free space is denoted by ϵ_0 (read as "epsilon naught").
- **Value:** The value of ϵ_0 is approximately: $\epsilon_0 = 8.85 \times 10^{-12}\ C^2/Nm^2$

Value of Coulomb's Constant (k)

By substituting the constant values of 4π and ϵ_0, Coulomb's constant 'k' can be calculated.

- **Value:** The approximate value of Coulomb's constant is: $k = 9 \times 10^9\ Nm^2/C^2$
- **Significance:** Coulomb's constant 'k' plays a role in Coulomb's Law similar to how the gravitational constant 'G' functions in Newton's Law of Gravitation ($F = G\frac{m_1 m_2}{r^2}$). It is a fundamental constant for electrostatic interactions.

Magnitude of Electrostatic Force

Understanding the Formula for Electrostatic Force

Remember that the formula for electrostatic force, often given as Coulomb's Law, provides only the **magnitude** of the force. It does not inherently give the direction.

$$F = k\frac{|q_1 q_2|}{r^2}$$

Where:

- F is the magnitude of the electrostatic force.
- k is Coulomb's constant.
- q_1 and q_2 are the magnitudes of the charges.
- r is the distance between the charges.

Significance of the Absolute Value Sign

The absolute value signs (||) around $q_1 q_2$ in the formula indicate that when calculating the magnitude of the force, we **only use the numerical values of the charges**, ignoring their positive or negative signs. The signs of the charges are crucial for determining the *direction* of the force (attractive or repulsive), but not its magnitude.

The Inverse Square Law

The formula shows an inverse square relationship with distance (r^2). While we can verify this relationship experimentally, the fundamental reason *why* it is r^2 and not r^3 or r^5 is a deeper question related to the nature of space and fields, which is not typically derived at this level.

Determining Force Direction

Once the magnitude of the force is calculated, the direction is determined by the nature of the charges involved:

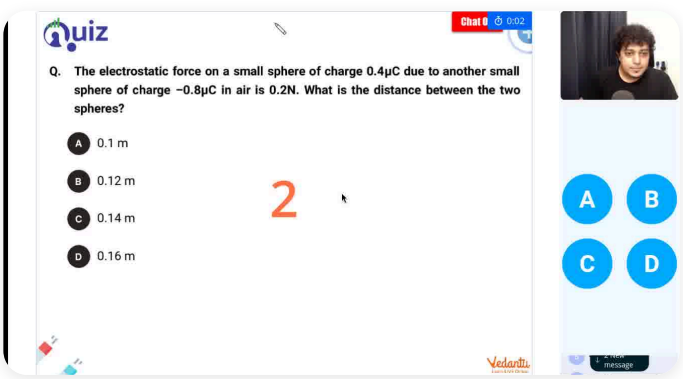
1. **Attractive Force (Opposite Charges):**
    - If one charge (q_1) is positive and the other (q_2) is negative, the force between them will be **attractive**.
    - This means each charge will pull the other towards itself.
    - **Example:** If charge A is positive and charge B is negative, charge A will attract charge B towards A, and charge B will attract charge A towards B. The forces act along the line connecting the two charges, pulling them closer.
2. **Repulsive Force (Like Charges):**
    - If both charges are positive (e.g., $+Q$ and $+Q$) or both are negative, the force between them will be **repulsive**.
    - This means each charge will push the other away from itself.
    - **Example:** If charge P is positive and charge Q is positive, charge P will push charge Q away from P, and charge Q will push charge P away from Q. The forces act along the line connecting the two charges, pushing them apart.

Newton's Third Law in Electrostatics

It's important to remember that electrostatic forces always obey Newton's Third Law of Motion. This means:

- The force exerted by q_1 on q_2 is equal in magnitude and opposite in direction to the force exerted by q_2 on q_1.
- For instance, if the attractive force on q_2 due to q_1 is 10 N, then the attractive force on q_1 due to q_2 will also be 10 N.

Electrostatic Force Problem



charge $-0.8\mu C$ in air is $0.2N$. What is the distance between the two spheres?

Given Information

- Charge of the first sphere (Q_1) = $0.4\mu C = 0.4 \times 10^{-6} C$
- Charge of the second sphere (Q_2) = $-0.8\mu C = -0.8 \times 10^{-6} C$
- Electrostatic force (F) = $0.2N$
- Medium: Air (implies Coulomb's constant $K = 9 \times 10^9 Nm^2/C^2$)

Objective

Find the distance (R) between the two spheres.

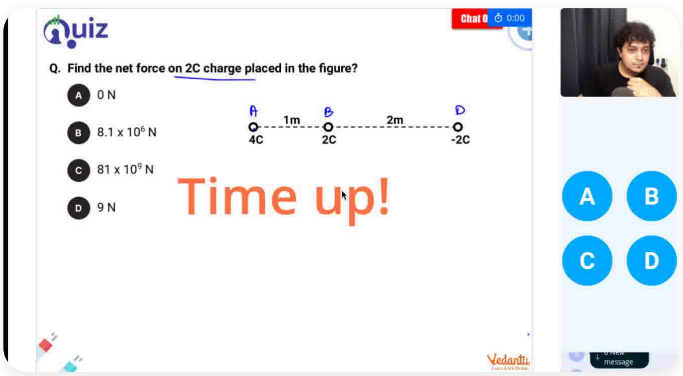
Solution Steps

1. **Identify the relevant formula:** Since we are dealing with electrostatic force between two point charges, Coulomb's Law is applicable: $F = K\frac{|Q_1 Q_2|}{R^2}$
2. **Rearrange the formula to solve for R:** $R^2 = K\frac{|Q_1 Q_2|}{F}$ $R = \sqrt{K\frac{|Q_1 Q_2|}{F}}$
3. **Substitute the given values into the formula:** $R^2 = \frac{(9\times10^9 Nm^2/C^2)\times|(0.4\times10^{-6} C)\times(-0.8\times10^{-6} C)|}{0.2N}$ Note: We use the magnitude of the charges, so the negative sign of Q_2 is dropped. $R^2 = \frac{9\times10^9\times(0.4\times10^{-6})\times(0.8\times10^{-6})}{0.2}$
4. **Simplify the expression:** $R^2 = \frac{9\times10^9\times0.32\times10^{-12}}{0.2}$ $R^2 = \frac{9\times0.32\times10^{-3}}{0.2}$ $R^2 = 9 \times 1.6 \times 10^{-3}$ $R^2 = 14.4 \times 10^{-3}$ To make it easier for square root, adjust powers of 10: $R^2 = 144 \times 10^{-4}$
5. **Calculate R by taking the square root:** $R = \sqrt{144 \times 10^{-4}}$ $R = \sqrt{144} \times \sqrt{10^{-4}}$ $R = 12 \times 10^{-2} m$ $R = 0.12m$

Final Answer

The distance between the two spheres is 0.12 m. This corresponds to option B.

Superposition of Forces



individual forces exerted by all other charges on that particular charge. This principle is known as the **superposition of forces**.

Problem Statement

Consider the following arrangement of charges:

- At point A: $+4$ C
- At point B: $+2$ C
- At point D: -2 C

These charges are arranged linearly with the following distances:

- Distance between A and B: 1 m
- Distance between B and D: 2 m

The objective is to find the net force on the $+2$ C charge located at point B.

Analysis of Forces

The $+2$ C charge at point B experiences two forces:

1. **Force due to charge at A (FA):** This is the force between the $+4$ C charge at A and the $+2$ C charge at B.
    - Since both charges are positive, the force will be **repulsive**. A will push B away from itself, meaning the force on B due to A will be directed towards D.
2. **Force due to charge at D (FD):** This is the force between the $+2$ C charge at B and the -2 C charge at D.
    - Since B is positive and D is negative, the force will be **attractive**. D will attract B towards itself, meaning the force on B due to D will also be directed towards D.

Calculation of Forces

We use Coulomb's Law to calculate the magnitude of each force:

$$F = \frac{k|q_1 q_2|}{r^2}$$

Where:

- k is Coulomb's constant (9×10^9 N m^2/C^2)
- q_1 and q_2 are the magnitudes of the charges
- r is the distance between the charges

- q_2 = +2 C (charge at B)
- r = 1 m (distance between A and B)

$F_A = \frac{k|(4)(2)|}{1^2} = \frac{k \cdot 8}{1} = 8k$ N

- Direction: Towards D (repulsive).

2. **Force by D on B (FD):**
   - q_1 = +2 C (charge at B)
   - q_2 = −2 C (charge at D)
   - r = 2 m (distance between B and D)

$F_D = \frac{k|(2)(-2)|}{2^2} = \frac{k \cdot 4}{4} = k$ N

- Direction: Towards D (attractive).

Net Force Calculation

Since both forces, F_A and F_D, are acting in the same direction (towards D), the net force on the +2 C charge will be the simple sum of their magnitudes.

$F_{\text{net}} = F_A + F_D$ $F_{\text{net}} = 8k + k = 9k$ N

Now, substitute the value of Coulomb's constant, $k = 9 \times 10^9$ N m^2/C^2:

$F_{\text{net}} = 9 \times (9 \times 10^9)$ N $F_{\text{net}} = 81 \times 10^9$ N

Conclusion

The net force on the +2 C charge is 81×10^9 N, directed towards D.

Important Note on Vector Sum

- If forces are in the **same direction**, their magnitudes are added to find the resultant.
- If forces are in the **opposite direction**, the magnitude of the smaller force is subtracted from the magnitude of the larger force to find the resultant. The direction of the resultant will be that of the larger force.
- If forces are at an angle, vector addition (e.g., parallelogram law or component method) must be used.

Session Wrap-up and Homework Assignment

Homework Assignment

Notes

Upcoming Topics

- The next session will cover more questions on Coulomb's law, forces, and equilibrium.
- Following this, the concept of the electric field will be introduced.

Next Class Details

- The next class is scheduled for Friday at 10 o'clock.

Did you find these notes helpful?